EXERCISE 8

Nine Qualitative Phase Portraits from Partial Data

Original statement · probls_ch4-ii_2627.pdf, p. 2

Qualitative Portraits • Classification • Nine Cases

STATEMENT

Draw the phase portrait for the equation $d\mathbf{x}/dt = A\mathbf{x}$ when the eigenvalues and eigenvectors of A are as given:

- (i) $\lambda_1 = 3$ with $\mathbf{v}_1 = (1, 1)$ and $\lambda_2 = -2$ with $\mathbf{v}_2 = (1, -2)$;
- (ii) complex conjugate eigenvalues $\lambda_{\pm} = -1 \pm 3i$, with $\dot{x} < 0$ when $x = 0$ and $y > 0$;
- (iii) a single eigenvalue $\lambda = 13$ with eigenvector $(3, 2)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$;
- (iv) $\lambda_1 = -2$ with $\mathbf{v}_1 = (2, 1)$ and $\lambda_2 = -3$ with $\mathbf{v}_2 = (1, -1)$;
- (v) a single eigenvalue $\lambda = -3$ with eigenvector $(1, -1)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$;
- (vi) $\lambda_{\pm} = \pm 2i$, where $\dot{y} < 0$ when $y = 0$ and $x > 0$;
- (vii) $\lambda_1 = 1$ with $\mathbf{v}_1 = (3, 2)$ and $\lambda_2 = 5$ with $\mathbf{v}_2 = (1, -4)$;
- (viii) $\lambda_{\pm} = 5 \pm i$, and $\dot{y} > 0$ when $y = 0$ and $x > 0$;
- (ix) a single eigenvalue $\lambda = -7$, with the matrix A being a multiple of the identity.

PHASE 1: SETUP: WHAT THE DATA DETERMINE

The data fix the **type** of the origin and the position of the eigenlines, which is all that the phase portrait of a linear system needs (Topic 4 Sec. 9, Robinson Ch. 31). A matrix consistent with the data is built in each item, to draw the portrait and to check the sign information numerically; any other matrix with the same data has the same portrait up to a linear distortion of the plane.

Data	Origin is a	Extra information used
two real eigenvalues, opposite signs	saddle	eigenlines; stable = negative eigenvalue
two real eigenvalues, same sign	node	tangent to the eigenvector of the smaller $\lvert\lambda\rvert$; stable iff $\lambda < 0$

Data	Origin is a	Extra information used
complex $\rho \pm i\omega$, $\rho \neq 0$	spiral	stable iff $\rho < 0$; sense from the sign of $\dot{x}$ at $(0,1)$ or $\dot{y}$ at $(1,0)$
imaginary $\pm i\omega$	centre	closed orbits; sense from the sign condition
single eigenvalue, one eigenvector	improper node	S-shaped orbits along the eigenline; side/orientation from the sign condition
single eigenvalue, $A = \lambda I$	star	straight rays

PHASE 2: FORMULATION: HOW THE SIGN CONDITIONS ARE READ

At the point $(0,1)$ the vector field is $A(0,1)^T = (a_{12}, a_{22})^T$, so the sign of $\dot{x}$ there is the sign of a_{12}: positive means motion to the right at the top of the plane, which is clockwise; negative means counter-clockwise. At $(1,0)$ the field is $(a_{11}, a_{21})^T$, so the sign of $\dot{y}$ there is the sign of a_{21}: negative means downwards on the right-hand side, i.e. clockwise; positive means counter-clockwise (Topic 4 Sec. 7.8).

METHOD OF THE CLASS NOTES [NOTES, PP. 9-10]

The classification of the notes decides each item. Two real eigenvalues (Case 1) give a node or a saddle, read from the sign table of λ_2/λ_1. A complex pair (Case 2) gives a spiral or a centre from the sign of ρ, with the sense of rotation read from the sign of $\dot{x}$ or $\dot{y}$ on an axis, as in the example of the notes. A repeated eigenvalue (Case 3) gives a star when $A = \lambda I$ and an improper node, with one eigenvector, otherwise.

PHASE 3: DERIVATION OF THE NINE PORTRAITS

(i) $\lambda_1 = 3$ with $\mathbf{v}_1 = (1,1)$ and $\lambda_2 = -2$ with $\mathbf{v}_2 = (1,-2)$

Two real eigenvalues with independent eigenvectors ($\det[\mathbf{v}_1\ \mathbf{v}_2] = -3 \neq 0$). A consistent matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} \frac{4}{3} & \frac{5}{3} \\ \frac{10}{3} & -\frac{1}{3} \end{pmatrix}$ (checked: $A\mathbf{v}_1 = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\mathbf{v}_1$, $A\mathbf{v}_2 = \begin{pmatrix} -2 \\ 4 \end{pmatrix} = -2\mathbf{v}_2$).

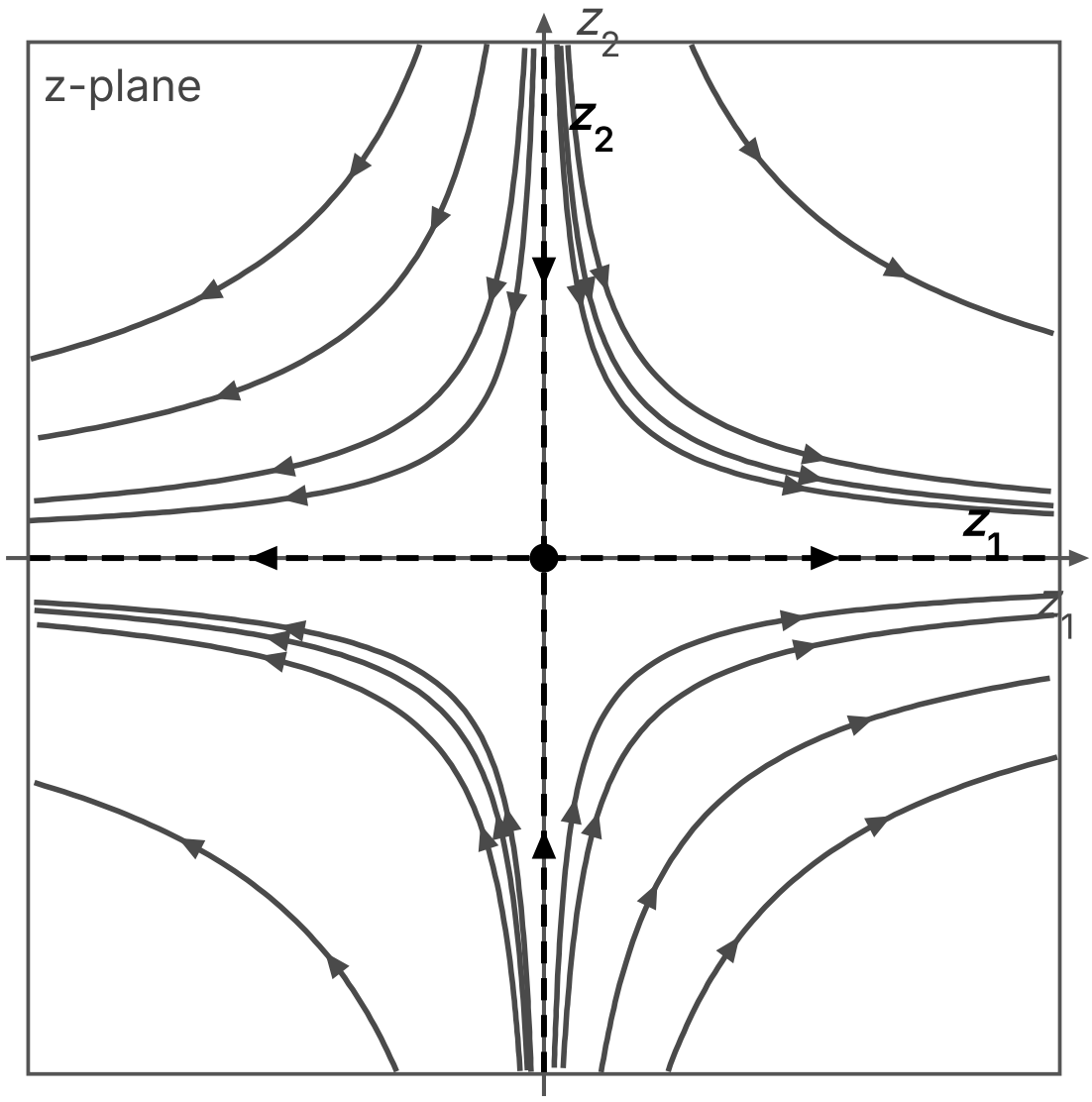
Phase portrait

$\lambda_1 = 3$ and $\lambda_2 = -2$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

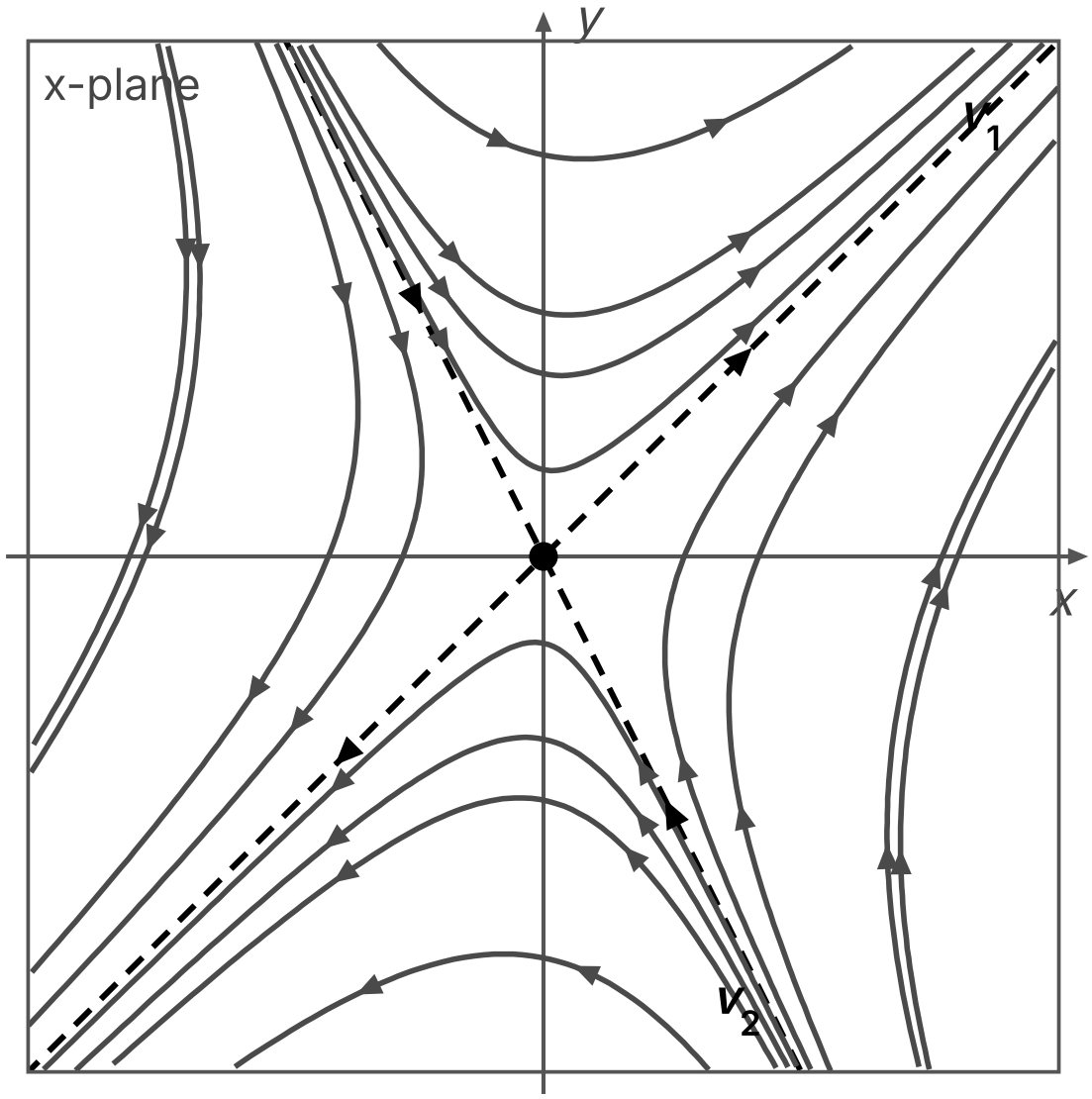
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{3t}$ and $z_2 = c_2 e^{-2t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \left(|z_1|/|c_1|\right)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{-\frac{2}{3}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = -\frac{2}{3}$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(1,-2)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(1,1)$ (positive eigenvalue) is the **unstable manifold**. All other orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -2)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 3\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -2\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-\frac{2}{3}}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -2)$.

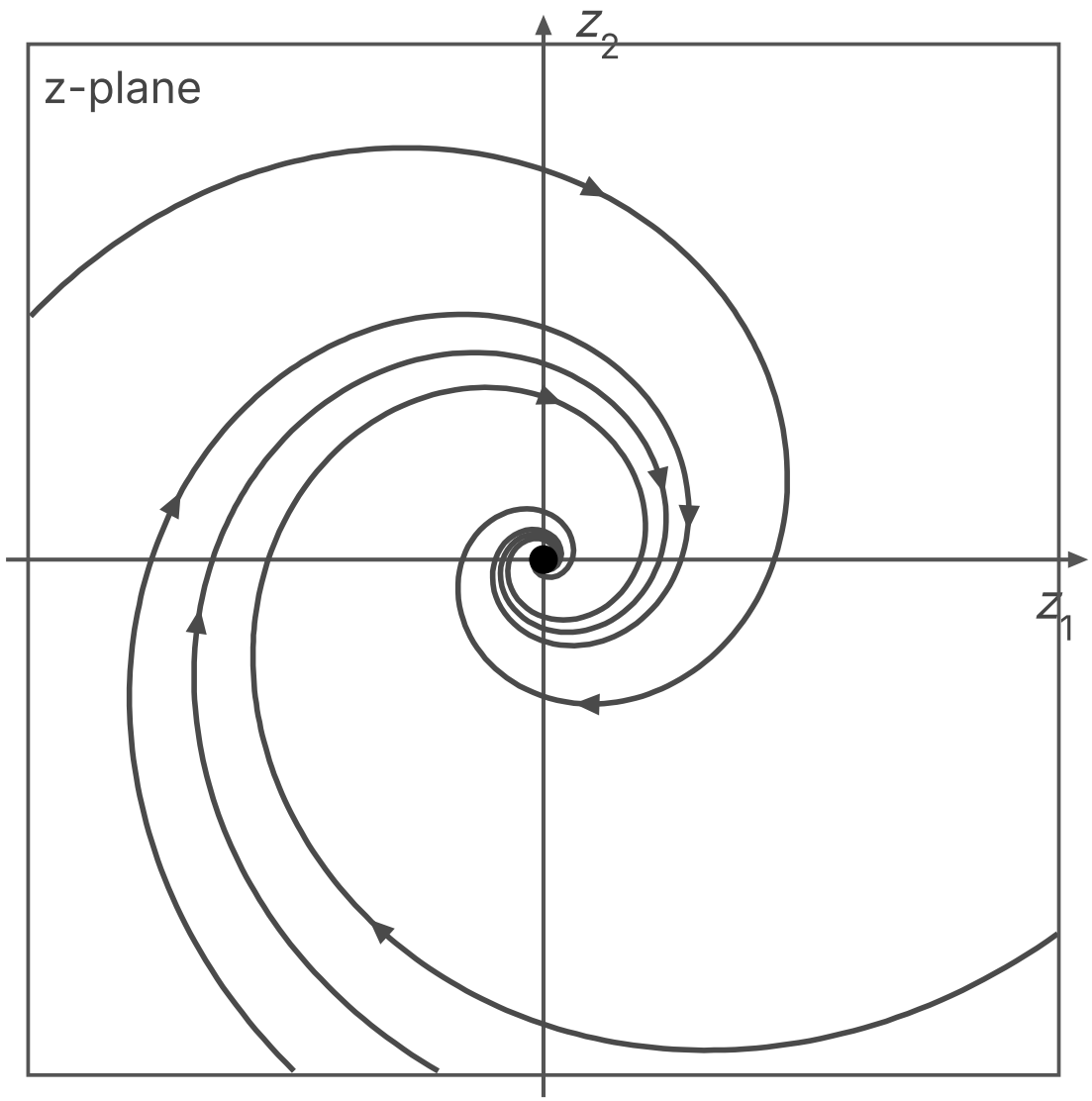
(ii) complex conjugate eigenvalues $\lambda_\pm = -1 \pm 3i$, with $\dot{x} < 0$ when $x = 0$ and $y > 0$

$\dot{x} < 0$ at $(0, 1)$ gives $a_{12} < 0$: leftwards at the top of the plane, i.e. **counter-clockwise**. A consistent matrix with eigenvalues $-1 \pm 3i$ is $A = \begin{pmatrix} -1 & -3 \\ 3 & -1 \end{pmatrix}$ ($\operatorname{tr} A = -2$, $\det A = 10$, $\lambda = -1 \pm 3i$; $a_{12} = -3 < 0$).

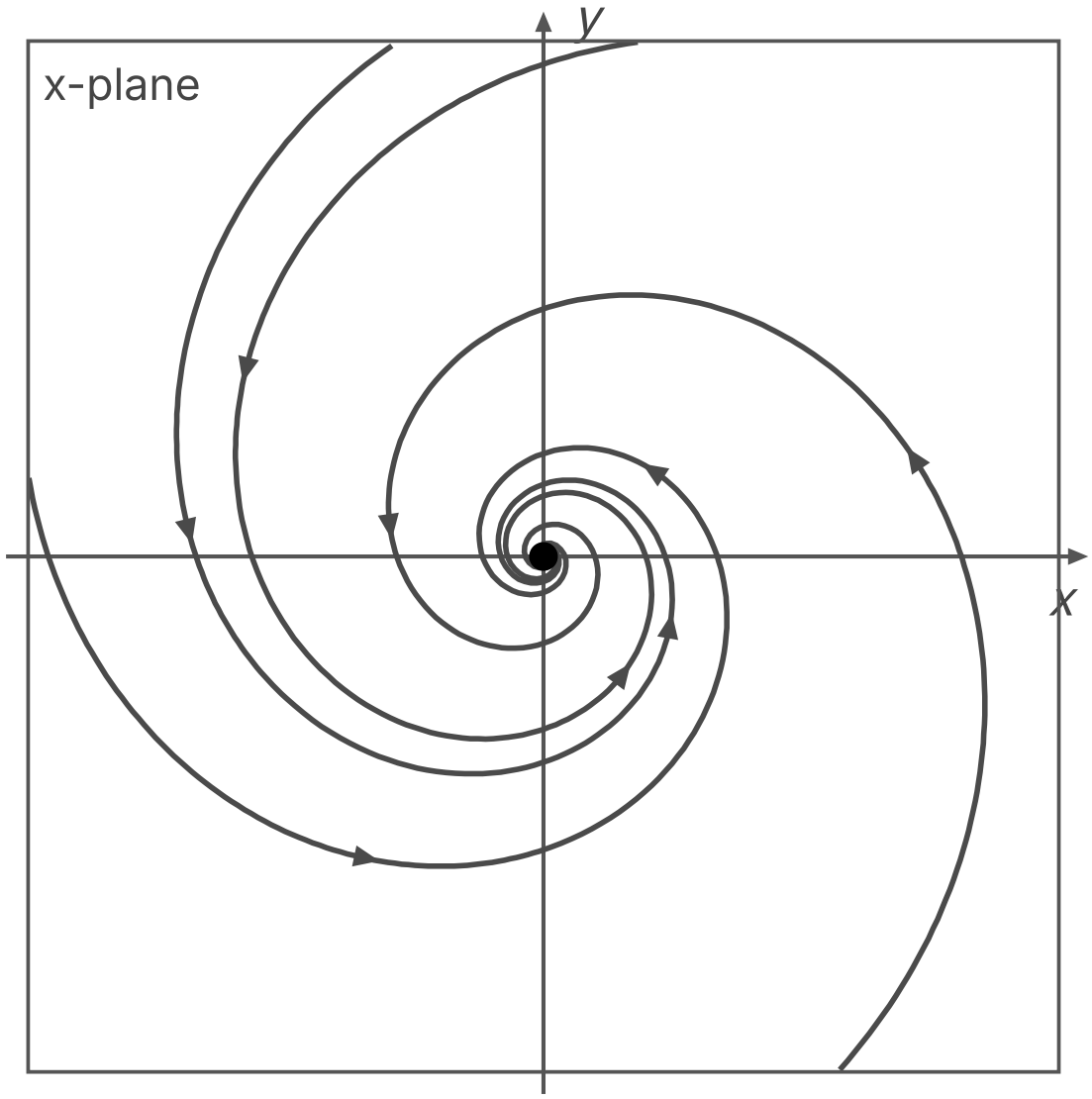
Phase portrait

$\rho = -1$ and $\omega = 3$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-1t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-\frac{2\pi}{3}}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$, so $\dot{x} = -3$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -9$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), counter-clockwise.

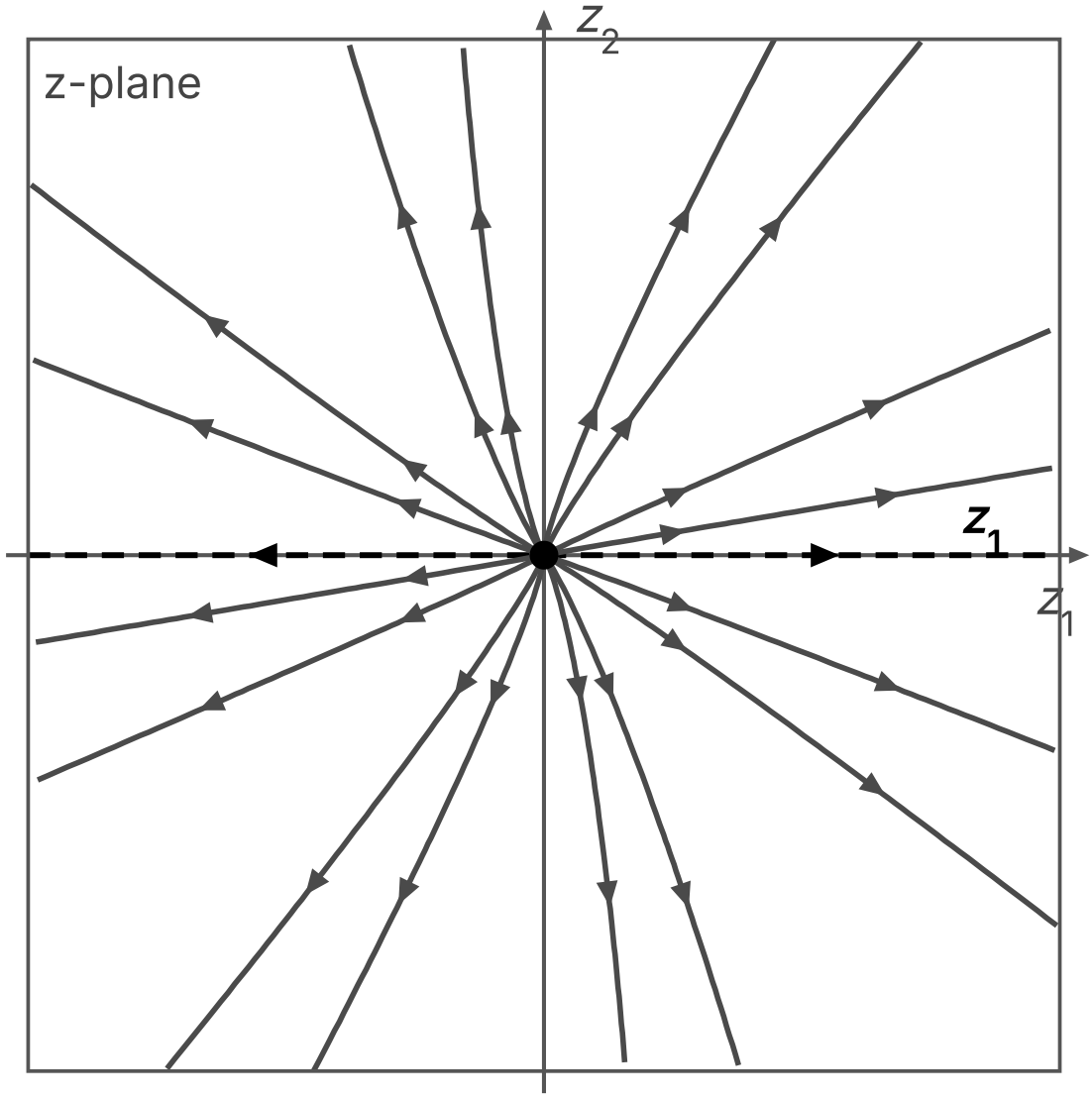
(iii) a single eigenvalue $\lambda = 13$ with eigenvector $(3, 2)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$

Single eigenvalue 13 with one eigenvector $(3, 2)^T$: $A \neq 13I$ (a star would have two independent eigenvectors), so $N = A - 13I$ is non-zero, $N^2 = O$ and $\ker N = \text{span}\{(3, 2)^T\}$. Hence $N = c\begin{pmatrix} 3 \\ 2 \end{pmatrix}(2, -3) = c\begin{pmatrix} 6 & -9 \\ 4 & -6 \end{pmatrix}$ (rows orthogonal to $(3, 2)$ so that $N(3, 2)^T = \mathbf{0}$), and $a_{12} = N_{12} = -9c$. The condition $\dot{x} > 0$ at $(0, 1)$ means $a_{12} > 0$, so $c < 0$; take $c = -1$: $N = \begin{pmatrix} -6 & 9 \\ -4 & 6 \end{pmatrix}$, $A = 13I + N = \begin{pmatrix} 7 & 9 \\ -4 & 19 \end{pmatrix}$.

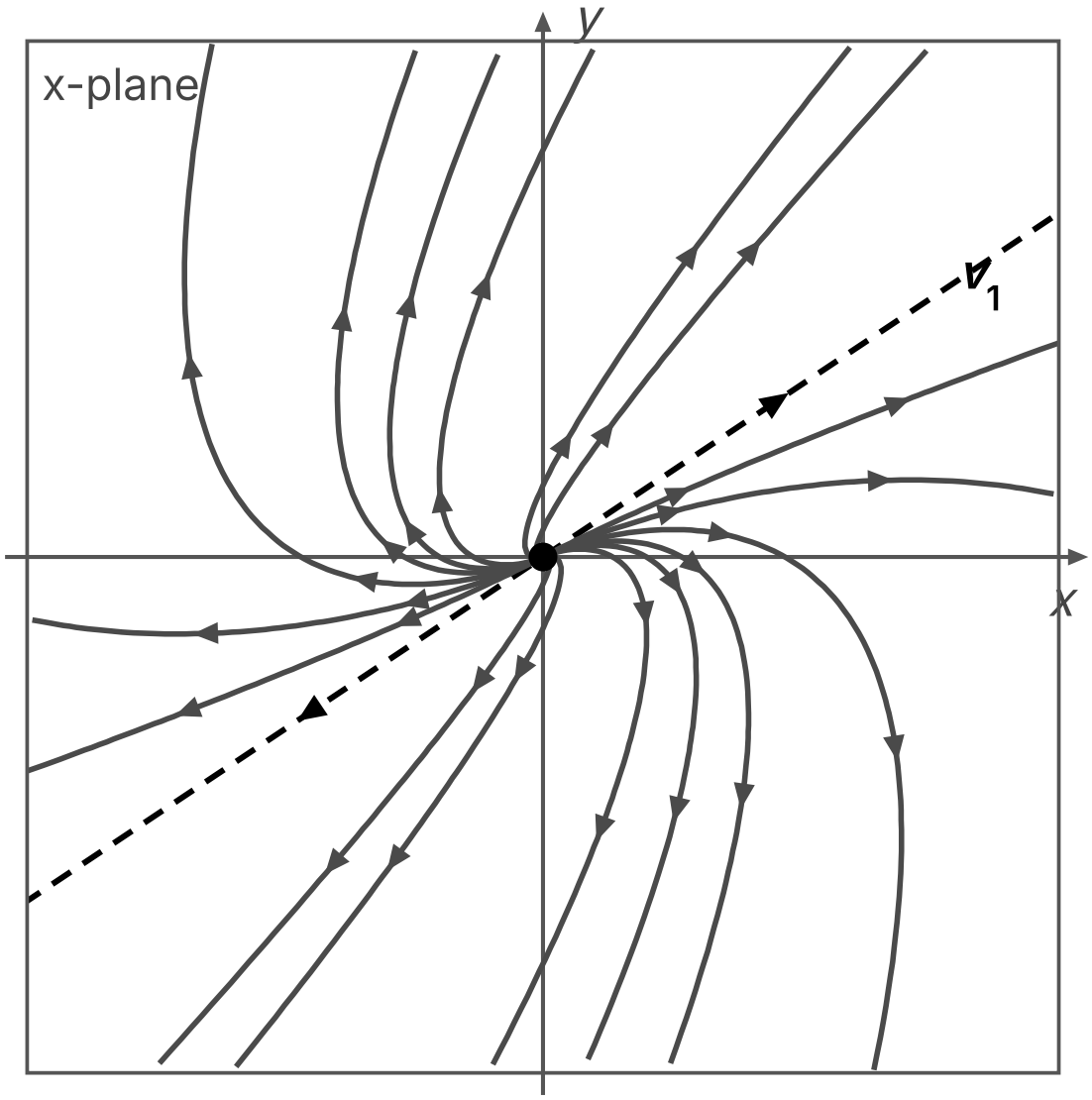
Phase portrait

$\lambda = 13$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **unstable improper node** (Topic 4 Sec. 8.8 and 8.9). In the $\mathbf{z}$-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{13t}$, arrows away from the origin); the other orbits satisfy $z_1 = z_2 \left[K + \frac{1}{\lambda} \ln |z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot{z}_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2 \mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix} 9 \\ 6 \end{pmatrix} = 3\,\mathbf{v}_1$, so there the orbit drifts along $+\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (unstable improper node).



Original (x, y)-plane (unstable improper node); dashed: the eigenline with $\mathbf{v}_1 = (3, 2)$.

(iv) $\lambda_1 = -2$ **with** $\mathbf{v}_1 = (2, 1)$ **and** $\lambda_2 = -3$ **with** $\mathbf{v}_2 = (1, -1)$

Two real eigenvalues with independent eigenvectors ($\det[\mathbf{v}_1\ \mathbf{v}_2] = -3 \neq 0$). A consistent matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} -\frac{7}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{8}{3} \end{pmatrix}$ (checked: $A\mathbf{v}_1 = \begin{pmatrix} -4 \\ -2 \end{pmatrix} = -2\mathbf{v}_1$, $A\mathbf{v}_2 = \begin{pmatrix} -3 \\ 3 \end{pmatrix} = -3\mathbf{v}_2$).

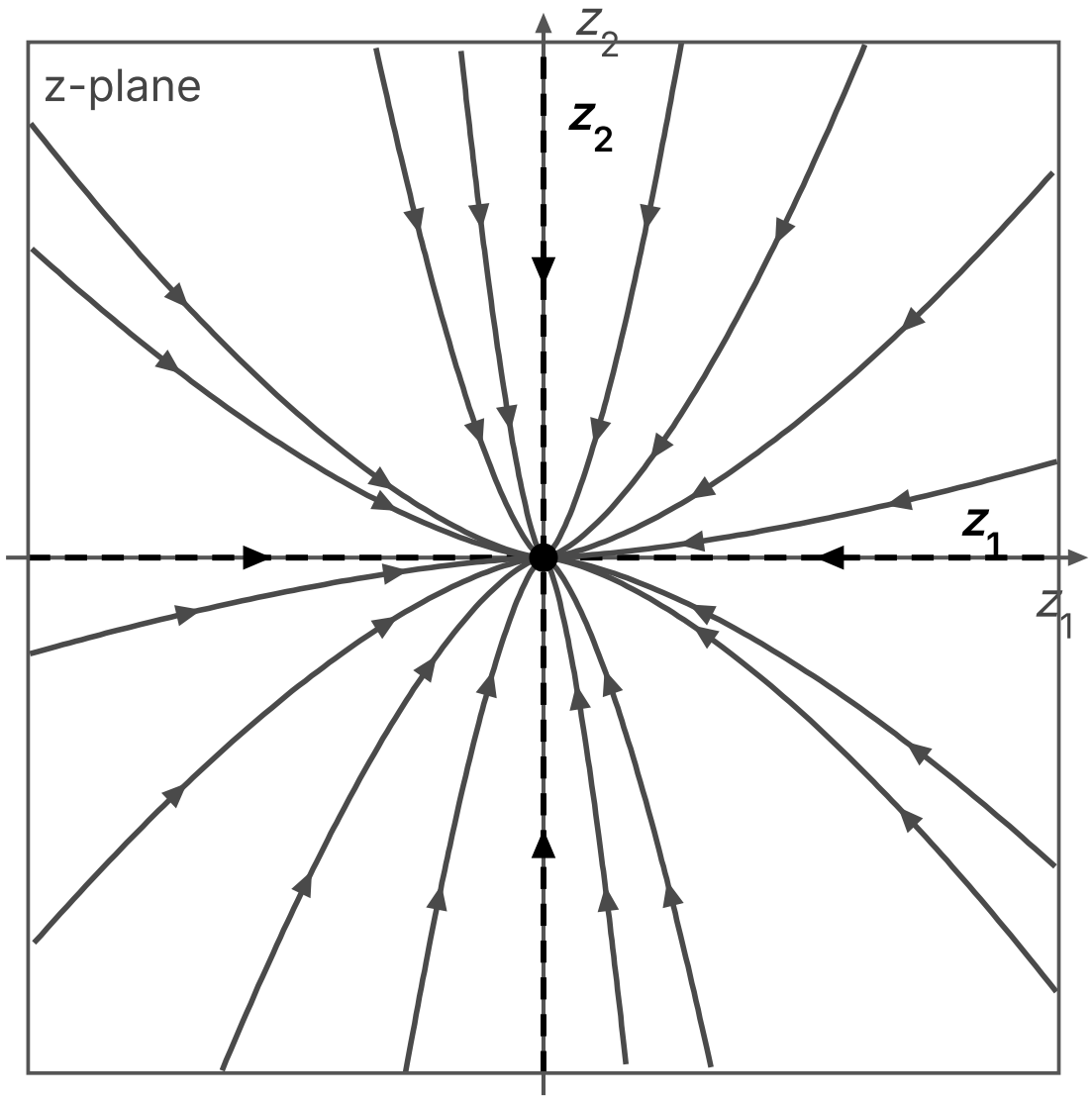
Phase portrait

$\lambda_1 = -2$ and $\lambda_2 = -3$ are both negative: the origin is a **stable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

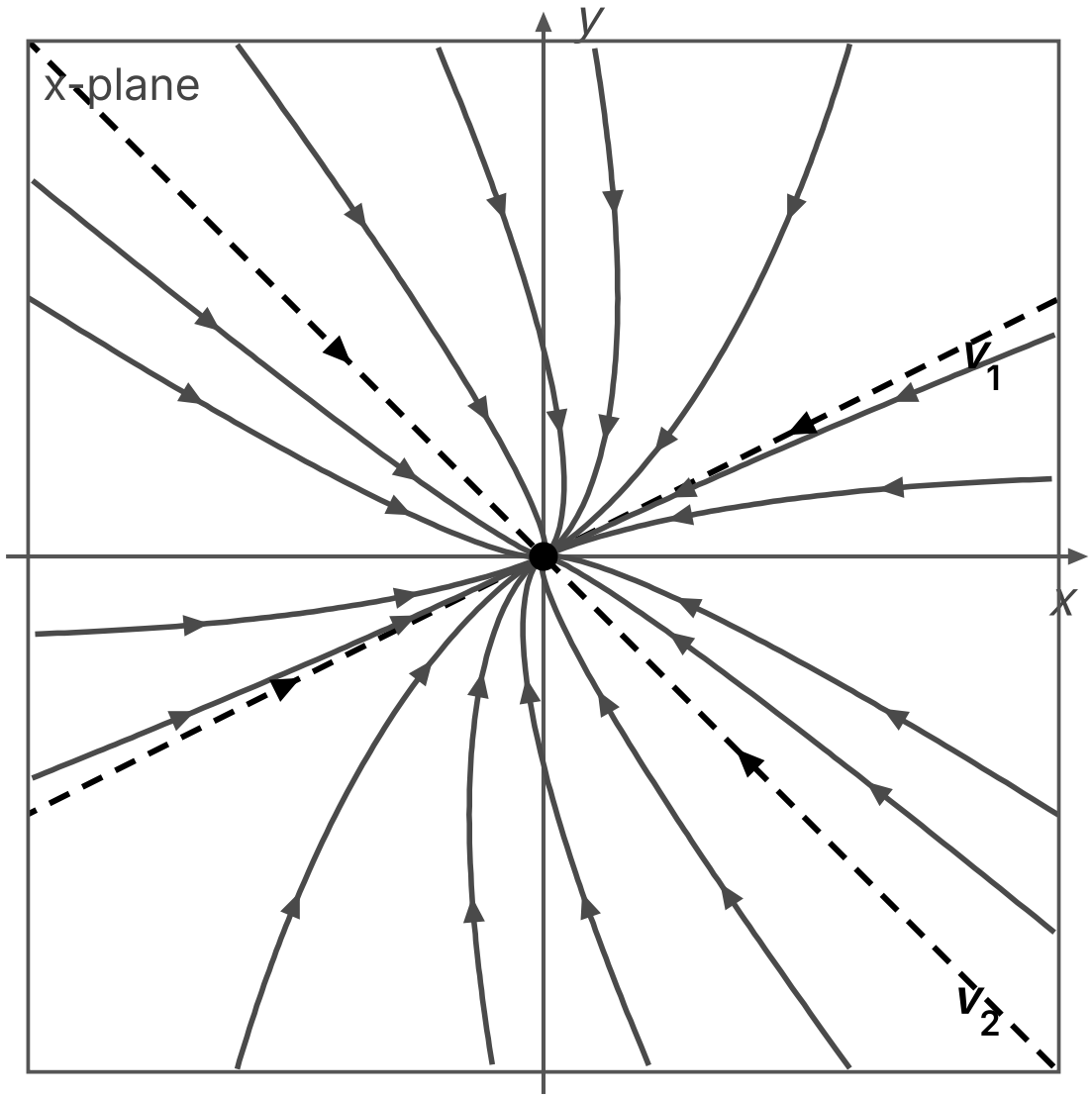
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{-2t}$ and $z_2 = c_2 e^{-3t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \left(|z_1|/|c_1|\right)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{\frac{3}{2}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = \frac{3}{2}$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 2$, i.e. to the line through $(2, 1)$ (they arrive there as $t \to +\infty$), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point towards the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (2, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = -2\mathbf{v}_1$ points inward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -3\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (stable node): invariant axes and orbits $z_2 = K|z_1|^{\frac{3}{2}}$.



Original (x, y)-plane (stable node): dashed lines are the eigenlines $\mathbf{v}_1 = (2, 1)$ and $\mathbf{v}_2 = (1, -1)$.

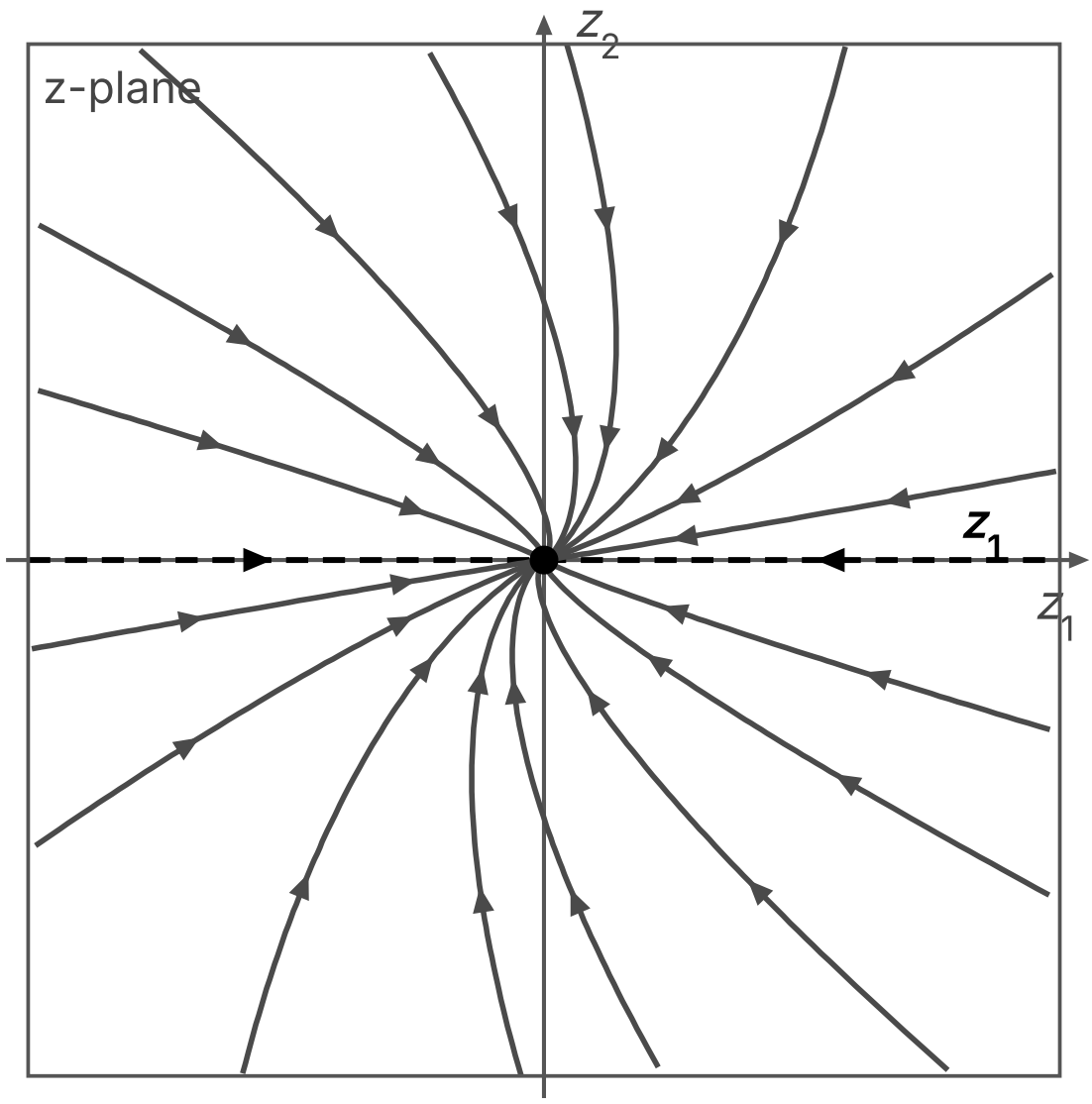
(v) a single eigenvalue $\lambda = -3$ with eigenvector $(1, -1)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$

Single eigenvalue -3 with one eigenvector $(1, -1)^T$: $A \neq -3I$ and $N = A + 3I \neq O$ with $\ker N = \text{span}\{(1, -1)^T\}$. Hence $N = c \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ and $a_{12} = N_{12} = c$. The condition $\dot{x} > 0$ at $(0, 1)$ means $a_{12} > 0$, so $c > 0$; take $c = 1$: $N = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$, $A = -3I + N = \begin{pmatrix} -2 & 1 \\ -1 & -4 \end{pmatrix}$.

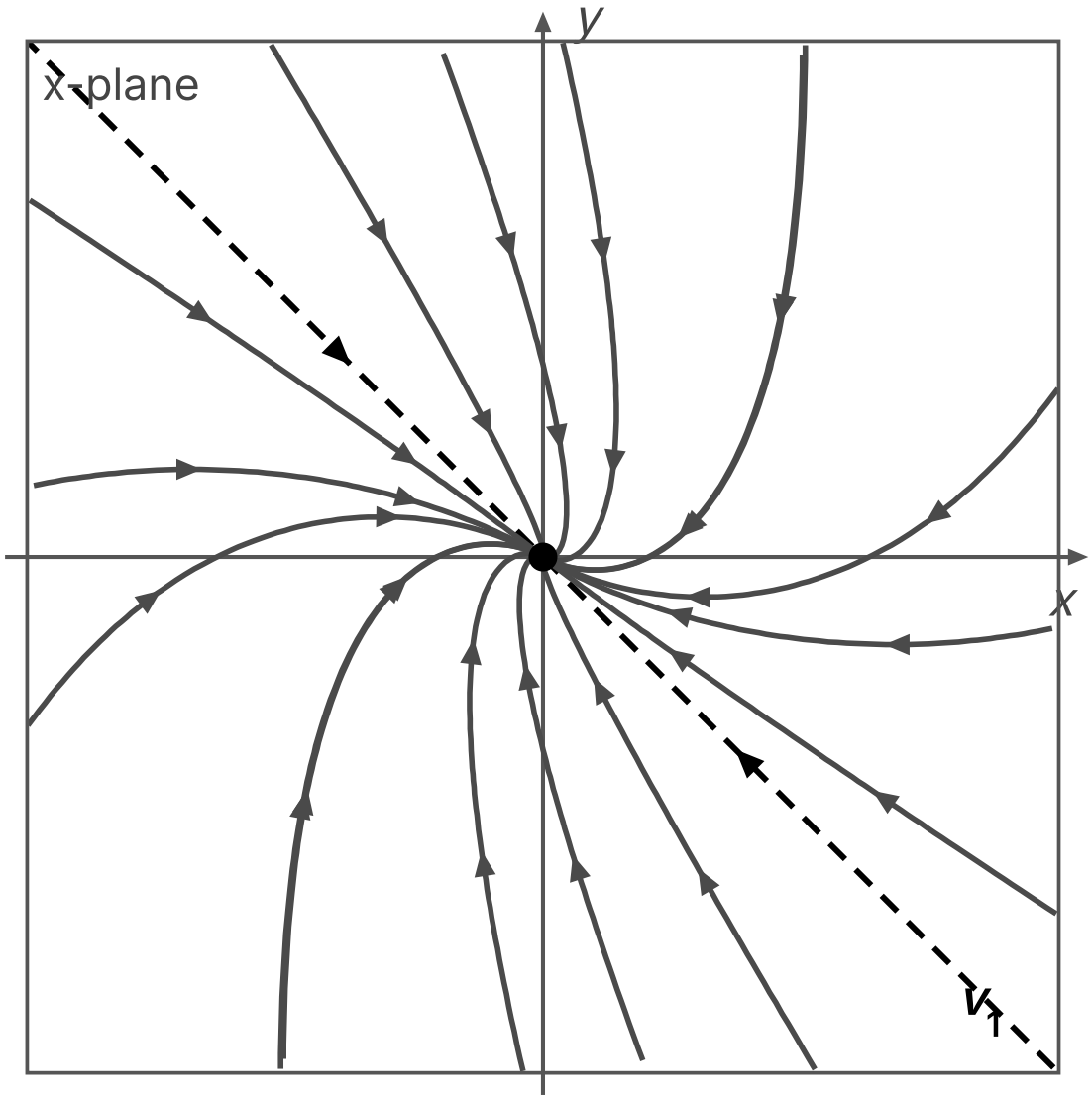
Phase portrait

$\lambda = -3$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **stable improper node** (Topic 4 Sec. 8.8 and 8.9). In the $\mathbf{z}$-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{-3t}$, arrows towards the origin); the other orbits satisfy $z_1 = z_2 \left[K + \frac{1}{\lambda} \ln |z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot{z}_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1\,\mathbf{v}_1$, so there the orbit drifts along $+\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (stable improper node).



Original (x, y)-plane (stable improper node); dashed: the eigenline with $\mathbf{v}_1 = (1, -1)$.

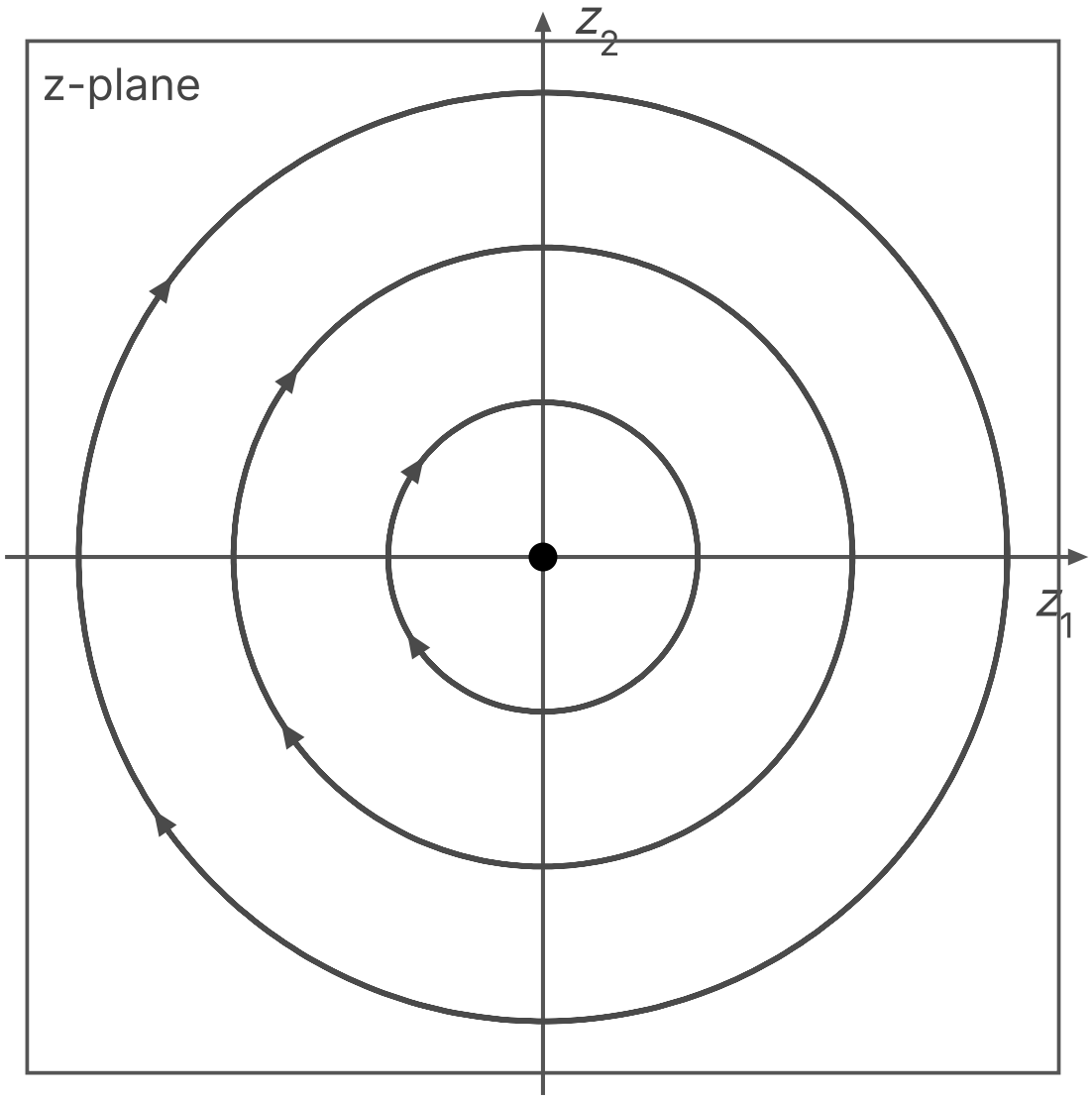
(vi) $\lambda_{\pm} = \pm 2i$, where $\dot{y} < 0$ when $y = 0$ and $x > 0$

$\dot{y} = a_{21}x + a_{22}y$ at $(1, 0)$ equals a_{21}, and $\dot{y} < 0$ there, so $a_{21} < 0$: at the right-hand side of the plane the motion is downwards, i.e. **clockwise**. A consistent matrix with eigenvalues $\pm 2i$ and $a_{21} < 0$ is $A = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ (tr $A = 0$, det $A = 4$, eigenvalues $\pm 2i$; $a_{21} = -2 < 0$).

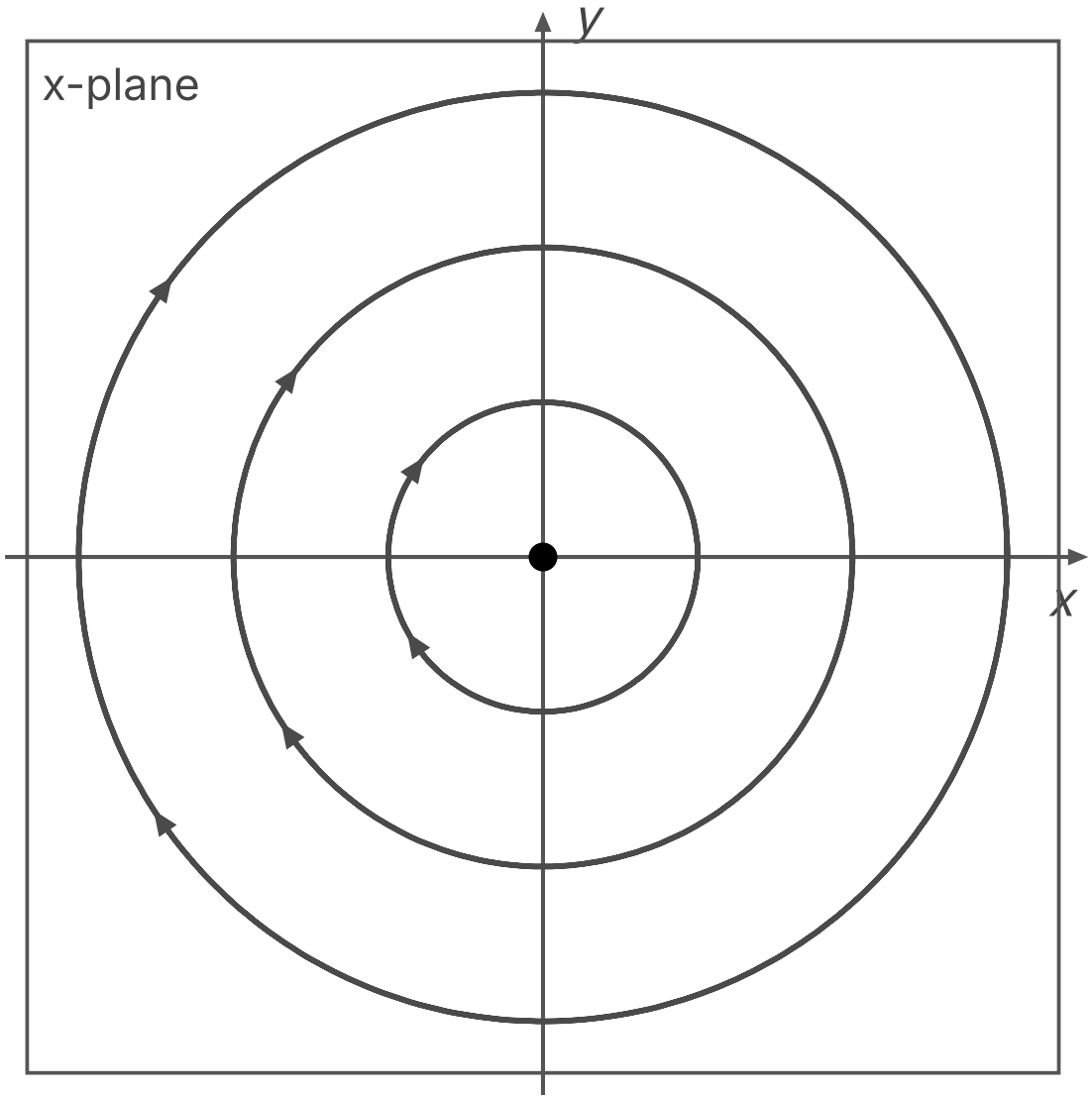
Phase portrait

$\rho = 0$ and $\omega = 2$: the origin is a **centre**. In the $\mathbf{z}$-plane $r = r_0 e^{0t}$ and $\theta = \theta_0 - 2t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and the orbits are circles of period $T = 2\pi/\omega$ (closed orbits).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0, 1)^T$, $\dot{\mathbf{x}} = A(0, 1)^T = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, so $\dot{x} = 2$ is > 0: to the right at the top of the plane: the rotation is **clockwise**.
Cross-check: det $Q = 4$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (centre): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (centre), clockwise.

(vii) $\lambda_1 = 1$ **with** $\mathbf{v}_1 = (3, 2)$ **and** $\lambda_2 = 5$ **with** $\mathbf{v}_2 = (1, -4)$

Two real eigenvalues with independent eigenvectors ($\det[\mathbf{v}_1\ \mathbf{v}_2] = -14 \neq 0$). A consistent matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} \frac{11}{7} & -\frac{6}{7} \\ -\frac{16}{7} & \frac{31}{7} \end{pmatrix}$ (checked: $A\mathbf{v}_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 1\mathbf{v}_1$, $A\mathbf{v}_2 = \begin{pmatrix} 5 \\ -20 \end{pmatrix} = 5\mathbf{v}_2$).

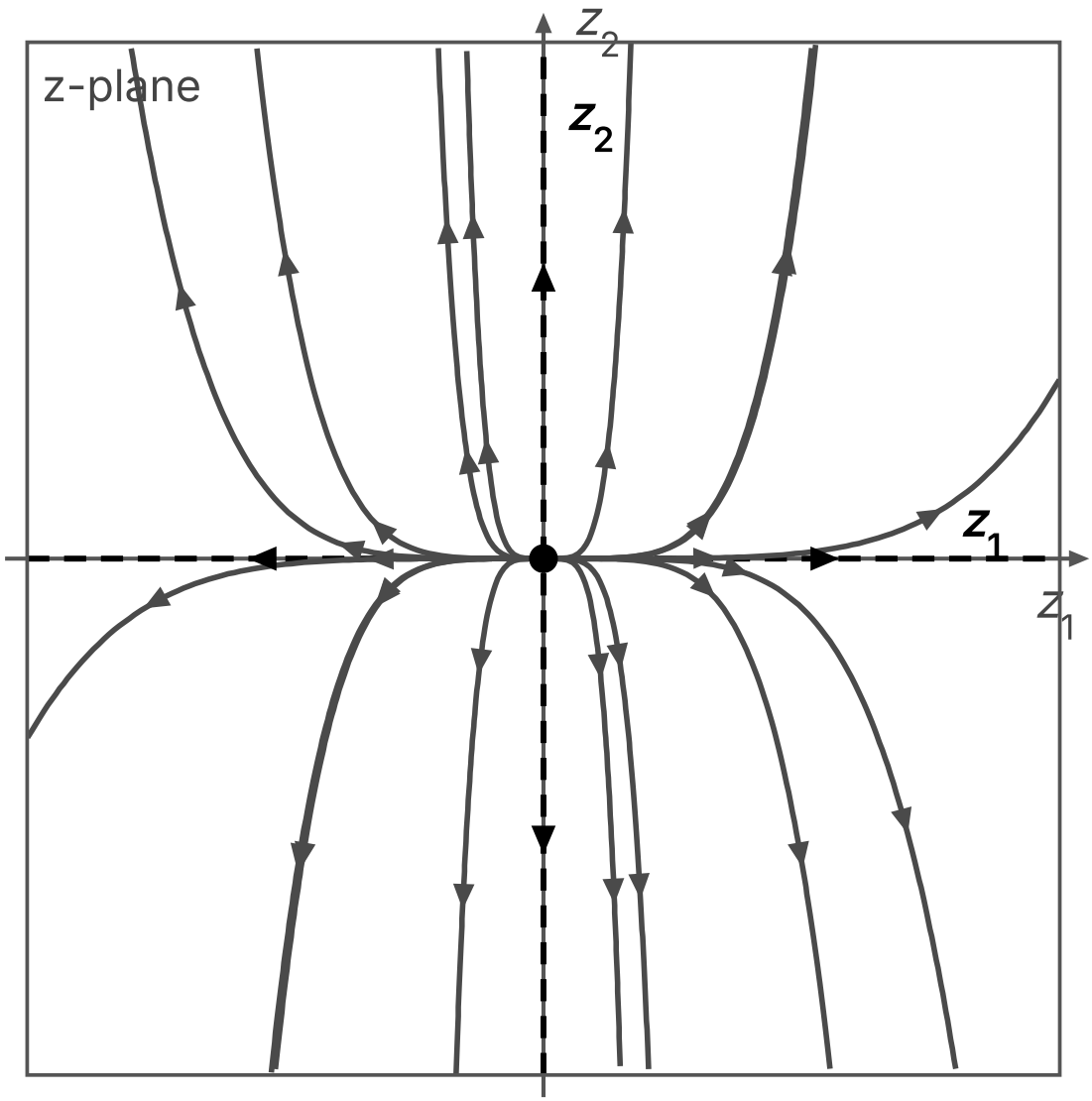
Phase portrait

$\lambda_1 = 1$ and $\lambda_2 = 5$ are both positive: the origin is a **unstable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

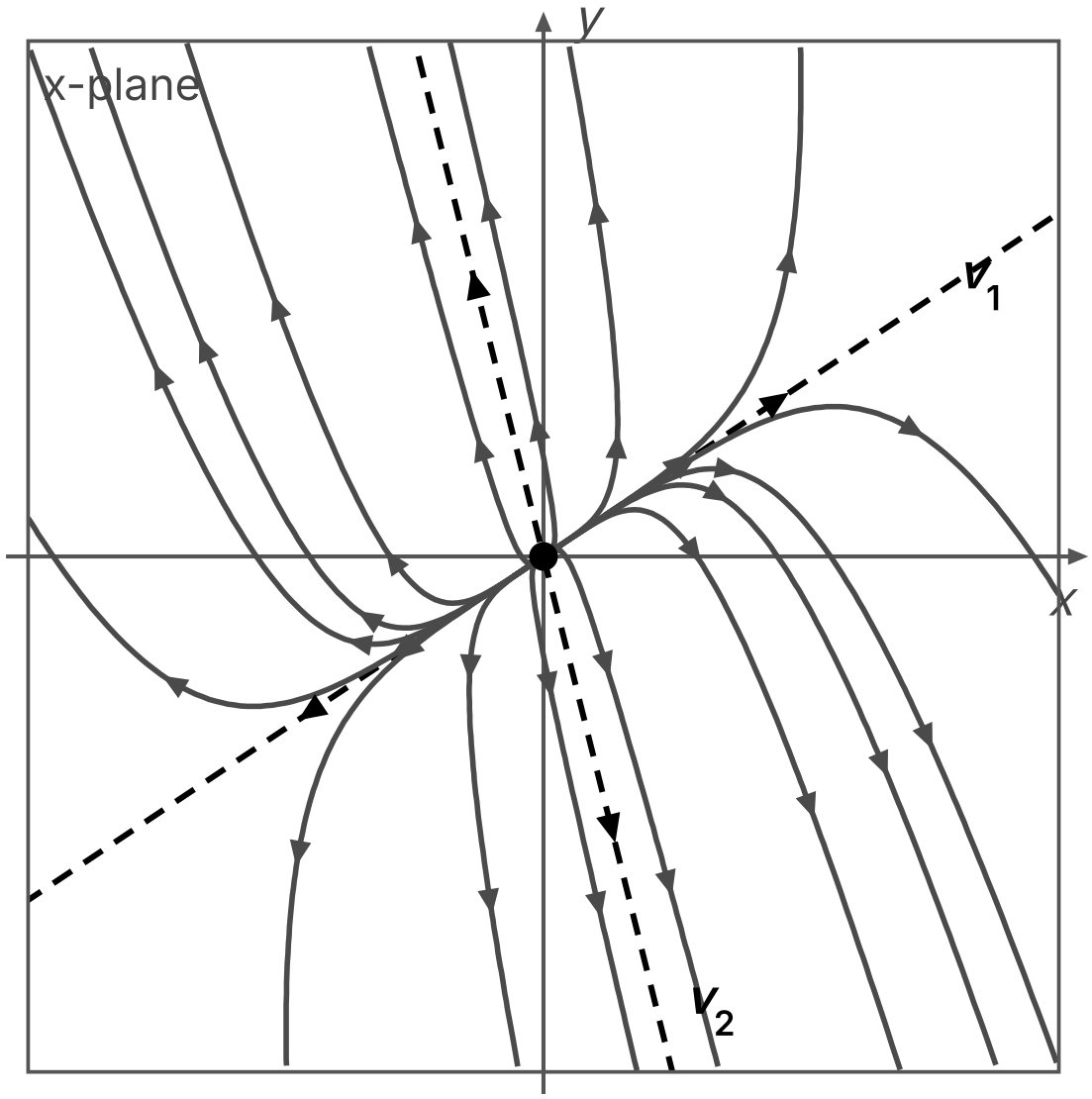
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{1t}$ and $z_2 = c_2 e^{5t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \left(|z_1|/|c_1|\right)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^5$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = 5$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 1$, i.e. to the line through $(3, 2)$ (they leave from there (the limit $t \to -\infty$)), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point away from the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (3, 2)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -4)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 1\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = 5\mathbf{v}_2$ points outward.



Canonical $\mathbf{z}$-plane (unstable node): invariant axes and orbits $z_2 = K|z_1|^5$.



Original (x, y)-plane (unstable node): dashed lines are the eigenlines $\mathbf{v}_1 = (3, 2)$ and $\mathbf{v}_2 = (1, -4)$.

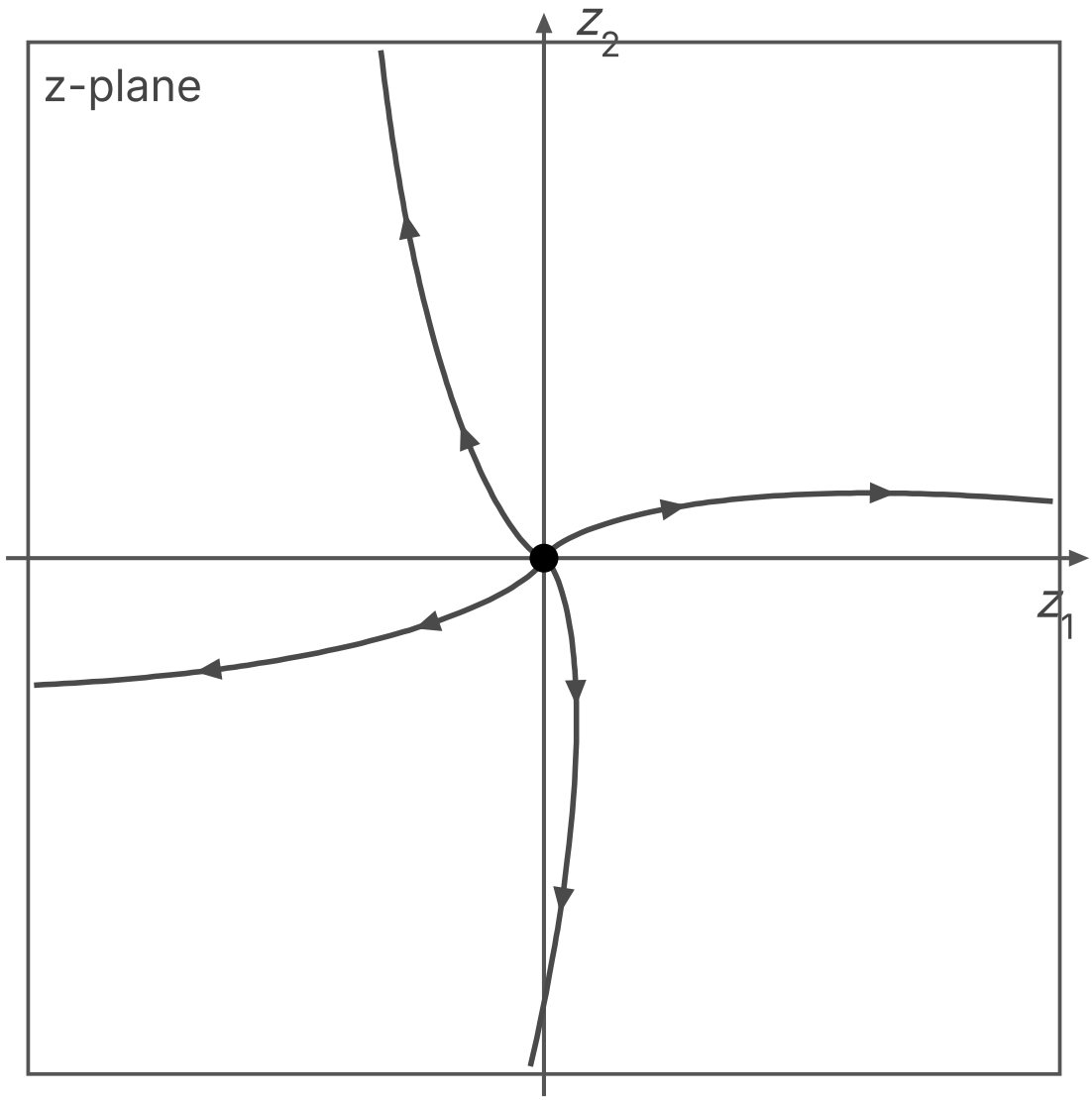
(viii) $\lambda_{\pm} = 5 \pm i$**, and** $\dot{y} > 0$ **when** $y = 0$ **and** $x > 0$

$\dot{y} > 0$ at $(1, 0)$ gives $a_{21} > 0$: upwards on the right-hand side, i.e. **counter-clockwise**. A consistent matrix with eigenvalues $5 \pm i$ is $A = \begin{pmatrix} 5 & -1 \\ 1 & 5 \end{pmatrix}$ (tr $A = 10$, det $A = 26$, $\lambda = 5 \pm i$; $a_{21} = 1 > 0$).

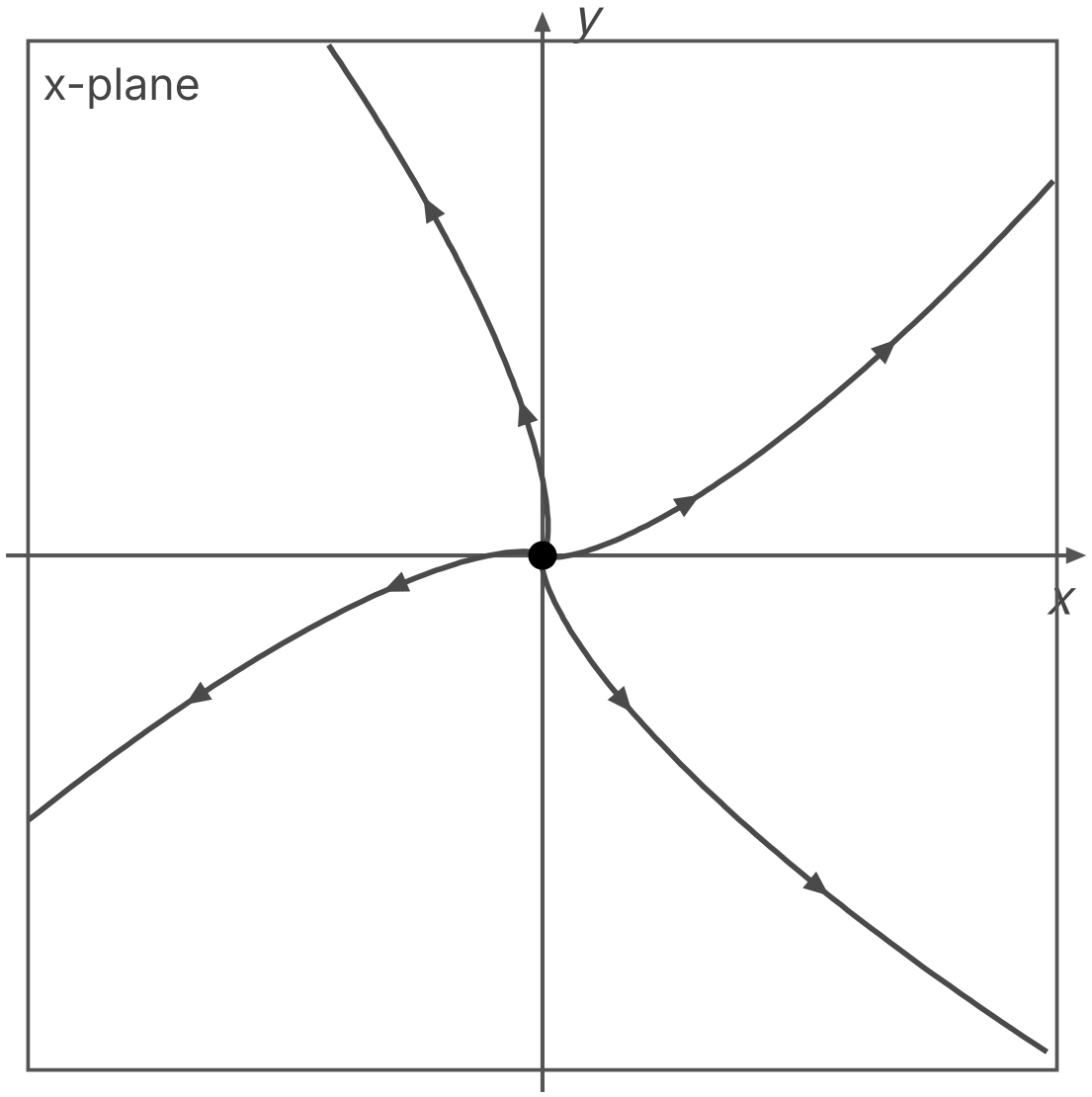
Phase portrait

$\rho = 5$ and $\omega = 1$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{5t}$ and $\theta = \theta_0 - 1t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{10\pi}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$, so $\dot{x} = -1$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -1$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot{\theta} = -\omega$.



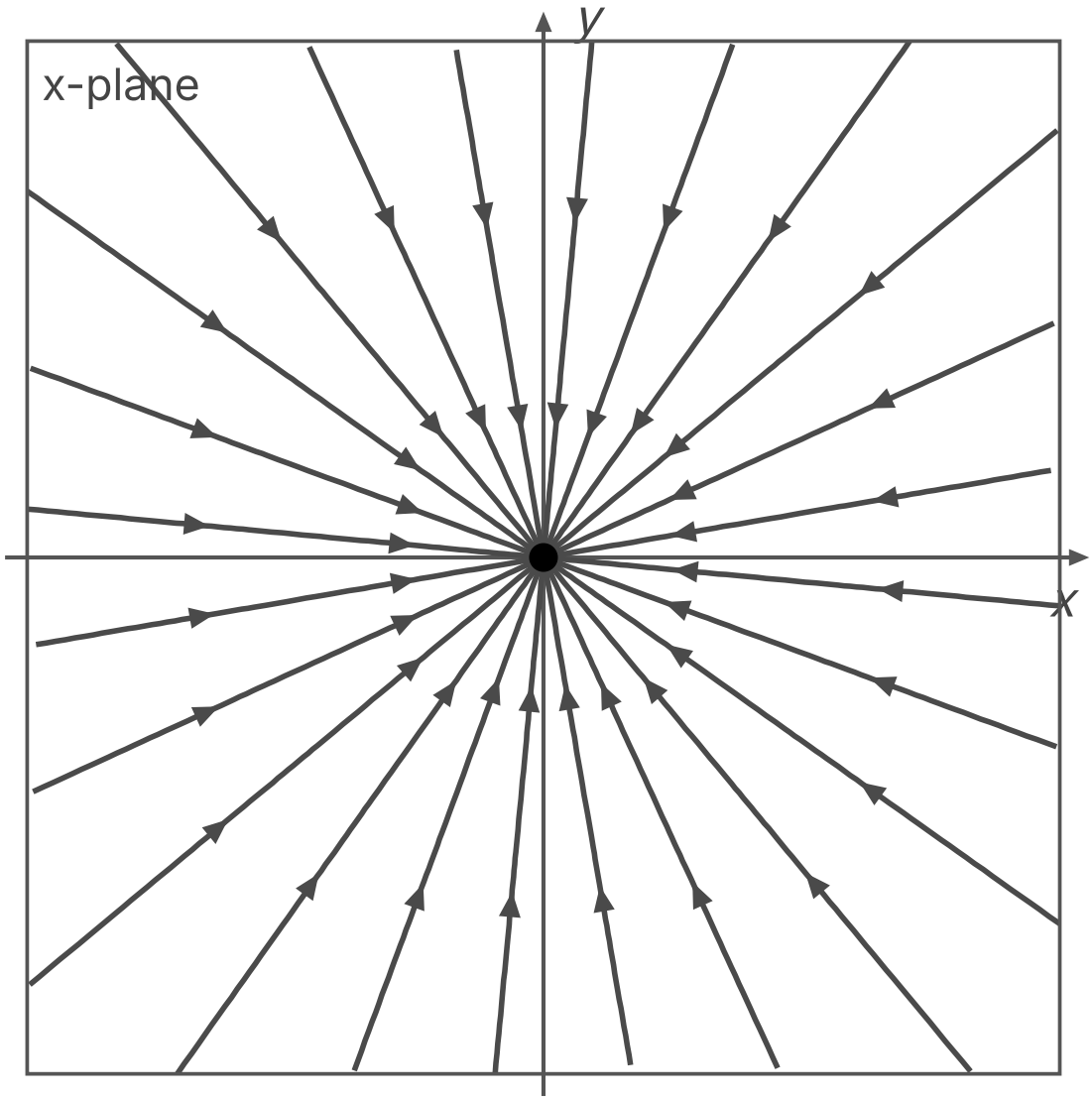
Original (x, y)-plane (unstable spiral), counter-clockwise.

(ix) a single eigenvalue $\lambda = -7$, with the matrix A being a multiple of the identity

A is a multiple of the identity with eigenvalue -7: $A = \begin{pmatrix} -7 & 0 \\ 0 & -7 \end{pmatrix}$.

Phase portrait

$A = -7I$: every line through the origin is invariant and every solution moves along its own ray, $\mathbf{x}(t) = e^{-7t}\mathbf{x}_0$, towards the origin (the origin is a **stable star**). The picture is the same in every basis, so there is no separate canonical plane.



Stable star: straight rays into the origin.

PHASE 4: INTERPRETATION AND CHECKS

Item	Origin	Stability
(i)	saddle	unstable
(ii)	spiral, counter-clockwise	stable
(iii)	improper node	unstable
(iv)	node, tangent to $\mathbf{v}_1 = (2, 1)$	stable
(v)	improper node	stable
(vi)	centre, clockwise	stable, not asymptotically
(vii)	node, tangent to $\mathbf{v}_1 = (3, 2)$	unstable

Item	Origin	Stability
(viii)	spiral, counter-clockwise	unstable
(ix)	star	asymptotically stable

- The nine items cover the nine pictures of Robinson's classification (Ch. 31, Fig. 31.1, as reproduced in Topic 4 Fig. 4.11; the unstable star is the time reversal of the stable one): saddle (i); stable node (iv); unstable node (vii); stable spiral (ii); unstable spiral (viii); centre (vi); stable star (ix); stable improper node (v); unstable improper node (iii).
- The tangency of nodes was verified on the numerical orbits: the orbits of (iv) are tangent at the origin to $\mathbf{v}_1 = (2, 1)$ ($|\lambda| = 2 < 3$) and those of (vii) to $\mathbf{v}_1 = (3, 2)$ ($1 < 5$).
- In (iii) and (v) the sign condition only chooses the orientation of the S-shaped orbits (on which side of the eigenline the orbit moves along $+\mathbf{v}_1$); the eigenline and the type do not depend on it.
- Trace-determinant check: (i) $\det A = -6 < 0$ saddle; (iv) $\mathrm{tr} = -5, \det = 6, \Delta = 1 > 0$ stable node; (vii) $\mathrm{tr} = 6, \det = 5$ unstable node; (iii) $\mathrm{tr}^2 = 4\det = 676$; (ix) $\mathrm{tr}^2 = 4\det = 196$.
- Spiral (viii) has $\rho/\omega = 5$: one turn multiplies the radius by $e^{2\pi\rho/\omega} = e^{10\pi} \approx 4 \times 10^{13}$, so it is a genuine spiral that, on the scale of the figure, looks like almost radial curves (the winding is invisible after the first fraction of a turn); in (ii), $\rho/\omega = -\frac{1}{3}$ the winding is clearly visible.